Visual Word Recognition and the Role of Diacritics: Evidence from Arabic Vowelization

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Across writing systems, diacritical marks on words provide phonological, morphological/lexical, and syntactic information. Despite their apparent importance, studies have revealed conflicting results regarding their influence on visual word recognition [1-3]. These findings have been interpreted as supporting two competing perspectives on the processing of diacritics: the *linguistic function hypothesis*, which proposes that diacritics are weighted according to their linguistic role in the writing system [1-2], and the *noisy-channel* (or *perceptual noise*) *model*, which views diacritics as sources of visual variability contributing little to lexical access [3]. Arabic provides a particularly informative test case for evaluating these perspectives. This is because in Arabic, vowelization signs mark short vowels and thus provide crucial phonological information; however, these diacritics are optional in most reading contexts. The present study therefore investigates the influence of these linguistically informative, yet optional diacritics on the earliest stages of Arabic visual word recognition.

Across three experiments, skilled adult Arabic readers completed a masked priming lexical decision task in which the primes and targets varied in terms of vowelization and morphological structure. Experiment 1 (*N*=56) tested unvowelized targets (محترف 'professional') preceded by three types of primes: unvowelized (محترف), correctly vowelized (مُحْتَرِفْ), and incorrectly vowelized (مَحُتِرَفِ). The unvowelized condition involved primes identical to the target. Under the correctly vowelized condition, the prime was the same as the target, but with the correct vowelization pattern. In contrast, the incorrectly vowelized condition preserved the letters but assigned non-corresponding diacritics. If vowelization diacritics are processed during the early stages of lexical processing, it was predicted that the incorrectly vowelized condition would yield a reduced priming effect. However, as indicated in Figure 1 (right panel), all three of these conditions yielded comparably robust priming effects. Experiment 2 (*N*=60) tested whether these effects hold even when targets are vowelized. To this end, this experiment tested vowelized targets (مُحْتَرِفْ 'professional') with the same three prime types as in Experiment 1: unvowelized (محترف), correctly vowelized (مُحْتَرِفْ), and incorrectly vowelized (مَحُتِرَفِ). In this case, the unvowelized primes had the same letters as the target but lacked any of its diacritical marks, while the correctly vowelized condition involved identity primes that corresponded exactly to the target. As indicated in Figure 2 (right panel), the results of this experiment were virtually identical to those of Experiment 1, with all three conditions yielding robust and comparable priming effects. Experiment 3 (*N*=51) extended this design to words lacking productive consonantal roots, such as loanwords (فيديو 'video') and underived nouns (طاولة 'table'). Such words do not conform to the typical tri-consonantal root structure of Arabic words and therefore provide a critical test of whether vowelization facilitates recognition in the absence of this morphological information. This experiment tested unvowelized targets (طاولة 'table') with two prime types: correctly vowelized (طَاوِلَةْ) and incorrectly vowelized (طَاوْلِةَ). As in the previous experiments, there were comparably robust priming effects under both priming conditions (Figure 3, right panel).

In sum, in these experiments, vowelization on the prime did not reliably modulate priming effects regardless of (i) whether the targets were vowelized/unvowelized or (ii) whether they involved productive tri-consonantal morphological roots. In fact, across all three experiments, correctly and incorrectly vowelized primes yielded robust priming effects of similar magnitude. This pattern of results indicates a clear insensitivity to information provided by these diacritics during Arabic lexical access. These findings thus demonstrate that the influence of diacritics relates to factors other than their linguistic function in the writing system. Indeed, although vowelization provide important phonological information, this information appears to be ignored during the early stages of Arabic visual word recognition. This is arguably the case because vowelization diacritics are optional in -- and often absent from -- Arabic texts. The presentation will detail the implications of these findings for models of diacritic processing.

Figure 1: Mean reaction times (left) and priming effects (right) across vowelization conditions in Experiment 1. Error bars represent 95% credible intervals.

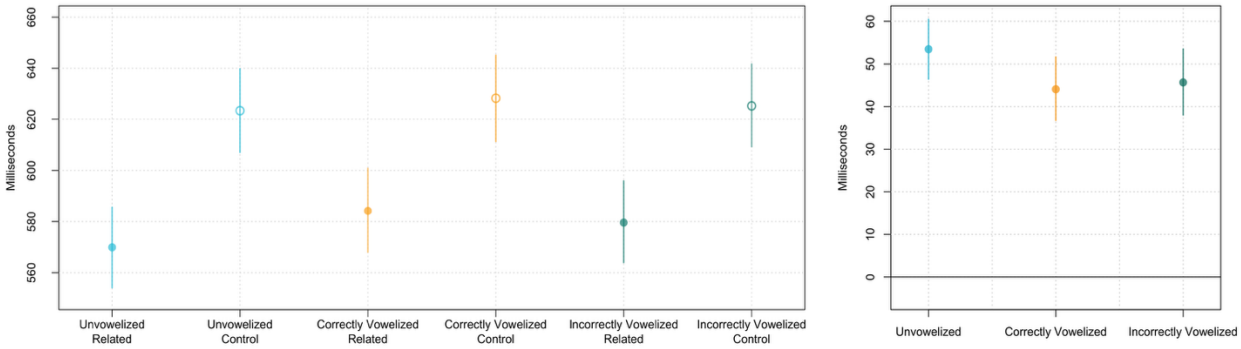


Figure 2: Mean reaction times (left) and priming effects (right) across vowelization conditions in Experiment 2. Error bars represent 95% credible intervals.

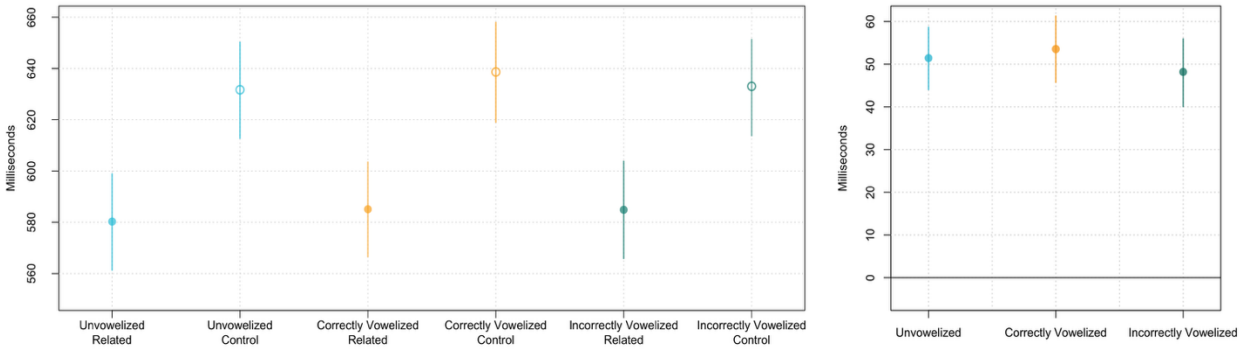
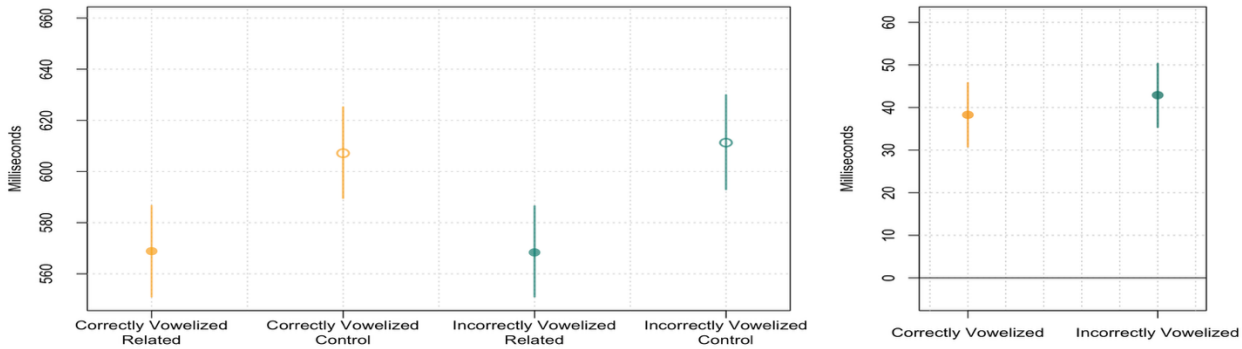


Figure 3: Mean reaction times (left) and priming effects (right) across vowelization conditions in Experiment 3. Error bars represent 95% credible intervals.



References

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